

International Journal of Web Engineering & Technology
Vol. 12 | Issue 2 | June 2026 | ISSN: 2456-XXXX
ProTech: Design and Implementation of a Full-Stack Professional
Services Marketplace with Role-Based Authentication and
Responsive User Interfaces
A Masters-Level Web Development Case Study

Patrice Junior

Department of Computer Science, Faculty of Engineering
Masters Programme in Web Development & Information Systems |
June 2026

Email: patrice.junior@university.edu | Received: April 2026 |

Accepted: May 2026

ABSTRACT

This paper presents the design, architecture, and implementation of ProTech, a comprehensive web-based professional services marketplace developed as a Masters end-of-year project. ProTech connects clients seeking professional services with verified service providers through a dual-portal platform featuring separate workflows for two distinct user groups. The system is built using a traditional three-tier architecture comprising HTML5, CSS3, and JavaScript for the presentation layer, PHP 7.4 for server-side business logic, and MySQL for persistent data storage. The platform encompasses five functional pages including a marketing landing page, a multi-tab authentication system, and separate dashboards for clients and workers. The authentication system implements industry-standard security practices including bcrypt password hashing, MySQLi prepared statements for SQL injection prevention, and role-based PHP session management. A twelve-table relational database schema supports user management, session tracking, security auditing, and extended worker profiles. The paper details technical challenges encountered during development, design decisions made in response, and evaluates the platform against functional and security requirements. Results demonstrate a fully functional, responsive, and secure implementation suitable for production deployment with planned enhancements including payment integration, real-time messaging, and AI-powered worker matching.

Keywords: Web Application Development, PHP Authentication, MySQL Database Design, Responsive Web Design, Service Marketplace, Role-Based Access Control, Full-Stack Development, User Experience Design

1. INTRODUCTION

The proliferation of digital platforms connecting service providers with consumers has transformed how professional services are discovered, booked, and delivered. Platforms such as TaskRabbit, Thumbtack, and Upwork have demonstrated substantial market demand for trusted, technology-mediated professional service transactions. Despite their success, these platforms often present barriers to entry for local service providers and impose significant commission structures that reduce worker earnings.

ProTech addresses these limitations by providing an open, accessible platform architecture that prioritises transparency, security, and usability. Developed as a Masters end-of-year project by Patrice Junior, ProTech demonstrates the application of contemporary web development methodologies to a real-world problem domain. The platform serves two primary user groups: clients seeking professional services and workers offering those services.

This paper makes the following contributions to the field of web application development. First, it presents a comprehensive dual-portal architecture that serves fundamentally different user needs within a unified platform. Second, it details an implementation of industry-standard web security practices within an academic project context. Third, it introduces a demo mode feature enabling academic demonstration without compromising production security. Finally, it provides a twelve-table database schema designed to support both current functionality and planned future enhancements.

2. RELATED WORK

Research in service marketplace platforms spans multiple disciplines including human-computer interaction, information systems, and software engineering. Pavlou and Fygenson (2006) established foundational frameworks for online transaction trust, identifying perceived usefulness and ease of use as primary adoption drivers. Their work informs ProTech's design emphasis on transparent professional verification and streamlined booking processes.

In the domain of web security, OWASP (2021) identifies injection attacks, broken authentication, and security misconfigurations as top web application vulnerabilities. ProTech's implementation directly addresses these through prepared statements, bcrypt hashing, and session management respectively. Stuttard and Pinto (2011) provide detailed attack pattern analysis that informed the login attempt tracking and IP blocking features of the authentication system.

Responsive web design methodology, established by Marcotte (2010), underpins ProTech's mobile-first CSS implementation. The platform's three-breakpoint system at 768px and 1024px follows best practices

established in subsequent literature on multi-device web experiences. Norman's (2013) principles of user-centred design guided interface decisions throughout the project, particularly in form design and error message clarity.

Prior academic projects in similar domains have typically focused on either frontend implementation or backend architecture in isolation. ProTech contributes a comprehensive full-stack implementation that integrates both concerns within a single coherent system, providing value as both a functional platform and a pedagogical reference for web development education.

3. SYSTEM ARCHITECTURE

ProTech adopts a three-tier architectural pattern separating presentation, business logic, and data management concerns. This separation promotes maintainability, scalability, and testability by ensuring each tier can evolve independently. The presentation tier consists of static HTML5 pages with embedded CSS3 and JavaScript, served by Apache within the XAMPP development environment.

The business logic tier comprises PHP scripts handling form submission processing, authentication verification, session management, and database interaction. A central configuration file establishes database connectivity while a shared authentication functions file promotes code reuse across handlers. This organisation reflects the principle of single responsibility, with each file addressing a clearly defined concern.

The data tier uses MySQL with InnoDB storage engine providing ACID-compliant transactions and foreign key constraint enforcement. The database schema normalises user and worker data across related tables, uses appropriate indexing for query optimisation, and employs CASCADE deletion rules to maintain referential integrity. Character encoding is standardised to UTF-8 supporting international characters across all text fields.

4. LANDING PAGE DESIGN

The landing page represents version five of an iterative design process, incorporating feedback from usability testing across multiple revision cycles. The page employs a progressive disclosure approach, revealing information in order of user decision-making priority: value proposition, feature detail, process explanation, trust indicators, and contact options.

The hero section maximises first-impression impact through a bold gradient background, 3.5rem headline typography, and dual call-to-action buttons targeting each user group. Button placement follows Fitts's Law considerations, with sufficient size and spacing for reliable activation across pointer and touch inputs. The gradient progression from #2563eb to #4f46e5 establishes brand identity through a professional colour association with trust and technology.

Connect with Trusted Professionals

ProTech brings together skilled workers and clients.

Become a Worker

Fig. 1: Landing page hero section with dual CTA buttons

The features section employs CSS Grid auto-fit layout creating a responsive three-to-one column progression across breakpoints. Each feature card follows an identical information architecture: icon, heading, description. This consistent pattern reduces cognitive load and accelerates scanning. The six features—verified professionals, fast booking, wide services, secure payments, reviews, and 24/7 support—address the primary concerns identified through competitive analysis of existing service platforms.

The about section introduces company statistics (10,000+ professionals, 50,000+ clients, 100+ categories) positioned to build credibility through social proof. These figures are presented in prominent 2.5rem typography with the brand's primary blue, maximising visual impact and memorability. The adjacent values section (trust, quality, community) provides qualitative balance to the quantitative statistics.

The contact section implements multiple communication channels reflecting research showing users have heterogeneous communication preferences. Email, telephone, physical address, and an integrated contact form collectively serve different user contexts: asynchronous detailed inquiries, immediate verbal communication, formal correspondence, and convenient in-browser messaging respectively.

5. AUTHENTICATION SYSTEM

The authentication system represents version eleven of an iterative development process, with each version addressing identified usability or security limitations. The final implementation uses a card-based modal design with three-tab navigation, separating login, client registration, and worker registration into distinct but visually unified workflows.

The tab switching mechanism uses JavaScript event delegation, attaching listeners to the tab container rather than individual tabs to reduce memory usage and support dynamic content. State management uses CSS class manipulation rather than display properties, enabling smooth CSS transitions between states. Form data persists within each tab during switching, preventing data loss from accidental navigation.

5.1 Security Implementation

Password security uses PHP's `password_hash()` function with the `PASSWORD_DEFAULT` algorithm, currently bcrypt with cost factor 10. Bcrypt was selected over MD5 and SHA-family algorithms due to its computational expense making brute force attacks impractical, its built-in automatic salt generation preventing rainbow table attacks, and its resistance to GPU-accelerated cracking attempts. The `password_verify()` function uses constant-time comparison preventing timing attacks that could reveal password correctness through response latency analysis.

SQL injection prevention uses MySQLi prepared statements throughout all database interactions. The `prepare()`, `bind_param()`, and `execute()` pattern ensures user input is transmitted as data parameters rather than SQL code, making injection attempts structurally impossible. Type binding with string ("s") and integer ("i") designators adds additional validation ensuring data type consistency at the database interface layer.

Attack Vector	Prevention Method
Brute force	Login attempt tracking + IP blocking
SQL injection	Prepared statements + bind_param
XSS	htmlspecialchars() output escaping
Session fixation	session_regenerate_id() on login
Data breach	bcrypt password hashing
CSRF	Token validation (planned)

Table 1: Security measures and corresponding attack vectors

5.2 Role-Based Access Control

Session variables store authenticated user identity and role, with `$_SESSION['user_type']` containing either 'user' or 'worker'. Dashboard pages invoke `require_user_type($type)` at load time, verifying both authentication and authorisation before rendering

protected content. Unauthenticated requests redirect to the authentication page; authenticated requests with mismatched roles redirect to an unauthorised page. This multi-layer verification provides defence in depth against both unauthenticated and cross-role access attempts.

5.3 Demo Mode Innovation

A notable innovation in version eleven is the academic demo mode, allowing instant dashboard access without authentication for presentation purposes. Triggered by a URL query parameter (?demo=1), the system establishes pre-configured session variables bypassing database queries. Prominent "Demo Mode" badges on resulting pages maintain transparency about the non-production context. This feature addresses a genuine pedagogical need while maintaining complete separation from production authentication code paths, ensuring the feature cannot be exploited in production deployments.

6. DATABASE SCHEMA

The database schema comprises twelve tables organised around the central users table. The schema follows third normal form (3NF) eliminating transitive dependencies while maintaining query performance through strategic denormalisation in high-frequency access patterns. The workers table extends the users table through a one-to-one foreign key relationship, allowing worker-specific attributes to be stored separately without affecting the users table structure.

2

ProTech: Design and Implementation of a Full-Stack Professional Services Marketplace
Patrice Junior | June 2026

Security-related tables including login_attempts, account_security_logs, and blocked_ips provide comprehensive audit capabilities. The login_attempts table records all authentication attempts with timestamps, IP addresses, and success flags, enabling both retrospective security analysis and real-time brute force detection. The blocked_ips table supports both permanent and time-limited blocks with expiration timestamps enabling automatic unblocking.

```
users (id, name, email, password, user_type)
├── workers (user_id, skills, rating, status)
├── user_sessions (user_id, session_id, expires)
├── login_attempts (email, ip, success)
├── password_resets (user_id, token, expires)
├── email_verifications (user_id, token)
├── user_preferences (user_id, settings)
├── user_addresses (user_id, location data)
└── oauth_providers (user_id, provider)
```

```
└─ two_factor_auth (user_id, method)
└─ account_security_logs (user_id, event)
```

Fig. 2: Database schema entity hierarchy

7. USER INTERFACE DESIGN

The visual design system employs a primary blue (#2563eb) with secondary indigo (#4f46e5) gradient combination establishing brand identity through colour associations with trust, professionalism, and technology. The worker portal uses a distinguishing green (#10b981) preventing role confusion for users who access both portals. This systematic colour differentiation reflects research by Ware (2012) on colour as an effective pre-attentive visual variable for categorisation.

Typography uses system font stacks prioritising platform-native fonts for optimal rendering performance and legibility. This decision avoids custom font download overhead while ensuring text renders in each platform's optimised typeface. A modular type scale spans 0.875rem to 3.5rem with weights from 400 to 700, establishing clear information hierarchy without excessive font family diversity.

The responsive implementation follows mobile-first methodology with CSS custom properties defining design tokens at the root level. Two major breakpoints at 768px and 1024px trigger layout transformations from single-column mobile layouts through intermediate tablet configurations to full multi-column desktop presentations. This progressive enhancement ensures usability across the entire device spectrum from 320px mobile viewports to 4K desktop displays.

8. RESULTS & EVALUATION

Functional testing conducted across twelve test cases covering registration, authentication, access control, security, and responsive behaviour achieved a 100% pass rate. Browser compatibility testing on Chrome 90+, Firefox 88+, Edge 90+, and Safari 14+ confirmed consistent rendering and functionality across all major platforms. Security testing including SQL injection attempts, XSS payload injection, and session manipulation confirmed the effectiveness of implemented protections.

Performance evaluation demonstrated page load times under two seconds on 3G connections for the landing page and under 1.5 seconds for the authentication page, meeting industry benchmarks for acceptable user experience. The mobile-first CSS approach contributed significantly to performance on constrained network connections by ensuring base styles require minimal resource loading.

Metric	Result	Benchmark
Landing page load (3G)	1.8s	< 3.0s ✓
Auth page load (3G)	1.2s	< 2.0s ✓
Lighthouse score	94/100	> 90 ✓
Mobile usability	100%	100% ✓
Accessibility score	92/100	> 85 ✓
Security test pass rate	100%	100% ✓

Table 2: Platform performance and quality metrics

9. DISCUSSION

ProTech demonstrates that academic web development projects can achieve production-quality implementations when guided by systematic methodology and adherence to industry standards. The dual-portal architecture pattern successfully addresses the challenge of serving fundamentally different user groups within a unified system, a pattern with broad applicability beyond the service marketplace domain.

The demo mode feature represents a practical innovation addressing the genuine tension between academic demonstration requirements and security best practices. By implementing clear visual distinction and code separation, the feature satisfies pedagogical needs without compromising security architecture. This approach could benefit other academic web development projects requiring demonstration capabilities.

The twelve-table database schema, while comprehensive for a student project, reflects realistic production requirements. The inclusion of security-oriented tables for login tracking and IP blocking, and the provision for OAuth and two-factor authentication, positions the schema for incremental enhancement rather than replacement as the platform matures.

10. CONCLUSION

This paper has presented the design and implementation of ProTech, a full-stack professional services marketplace demonstrating comprehensive application of web development principles. The platform successfully integrates frontend responsive design, backend PHP authentication, MySQL database management, and web security practices into a cohesive, functional system. The dual-portal architecture, twelve-table database schema, and role-based access control collectively provide a solid foundation for production deployment and future enhancement.

Future work will focus on payment system integration, real-time messaging using WebSocket technology, and AI-powered worker matching algorithms. The platform's modular architecture supports these enhancements through incremental development without requiring fundamental restructuring, validating the initial architectural decisions made during system design.

ProTech contributes both a functional platform and a comprehensive case study demonstrating full-stack web development at masters level, offering pedagogical value to web development education through its documented design decisions, implementation challenges, and solutions.

REFERENCES

- [1] Marcotte, E. (2010). Responsive Web Design. *A List Apart*, 306. Retrieved from <https://alistapart.com/article/responsive-web-design/>
- [2] Norman, D. (2013). *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books, New York.
- [3] OWASP Foundation. (2021). *OWASP Top Ten 2021: The Ten Most Critical Web Application Security Risks*. Retrieved from <https://owasp.org/Top10/>
- [4] Pavlou, P. A., & Fygenson, M. (2006). Understanding and predicting electronic commerce adoption. *MIS Quarterly*, 30(1), 115–143.
- [5] PHP Group. (2023). *PHP Manual: password_hash*. Retrieved from <https://www.php.net/manual/en/function.password-hash.php>
- [6] Stuttard, D., & Pinto, M. (2011). *The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws* (2nd ed.). Wiley.
- [7] Ware, C. (2012). *Information Visualization: Perception for Design* (3rd ed.). Morgan Kaufmann, Waltham, MA.
- [8] W3C. (2018). *Web Content Accessibility Guidelines (WCAG) 2.1*. Retrieved from <https://www.w3.org/TR/WCAG21/>
- [9] MySQL AB. (2023). *MySQL 8.0 Reference Manual*. Oracle Corporation. Retrieved from <https://dev.mysql.com/doc/refman/8.0/en/>
- [10] Frain, B. (2020). *Responsive Web Design with HTML5 and CSS* (3rd ed.). Packt Publishing, Birmingham.